

Work on Your Terms? Autonomy, Satisfaction, and Well-Being in Nonstandard Employment

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Abstract

This thesis investigates the relationship between alternative work arrangements (AWAs) and subjective well-being using cross-sectional data from the General Social Survey (2006–2022). AWAs include independent contractors/consultants/freelancers, contract firm employees, and on-call workers.

Through conducting multivariate regression and mediation analysis, independent contractors report significantly higher job satisfaction and modestly greater happiness than regular employees. These well-being advantages are largely mediated by job satisfaction, suggesting autonomy and preference alignment as key drivers. On-call workers report significantly lower job satisfaction without a corresponding reduction in happiness, reflecting a suppression effect. Contract firm employees show no significant differences in either domain. Across fully adjusted OLS models, ordered logistic regressions, and robustness checks with industry-clustered errors, income and marital status consistently predict higher well-being. Professional status strongly predicts job satisfaction, though its association with happiness weakens when job satisfaction is accounted for. These findings highlight the heterogeneity within nonstandard employment and suggest that job satisfaction, more than happiness, should be treated as the primary lens for understanding how employment structure shapes well-being.

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1. Introduction

“Be your own boss, decide when and where you work, and when you take breaks – all while earning reliable cash.”

This slogan, featured on Lyft’s website, encapsulates the broader marketing appeal of the gig economy, which promotes independence, flexibility, and financial control. Gig work, short-term, task-based labor arranged through digital platforms like Uber, Lyft, and Upwork, has become a widely recognized form of nontraditional employment. These platforms connect workers with on-demand tasks, allowing them to engage in income-generating activity without the constraints of a traditional employer-employee relationship (Wood et al., 2018).

However, gig work represents only one visible corner of a much broader landscape of nonstandard labor arrangements. The Bureau of Labor Statistics defines these as Alternative Work Arrangements (AWAs), encompassing four main categories: independent contractors/consultants/freelancers, on-call workers, contract firm employees, and temporary agency staff. For simplicity, this study refers to “independent contractors/consultants/freelancers” collectively as independent contractors. While many gig workers fall into this category, not all independent contractors participate in platform-based labor like Lyft or Uber. Some contractors instead engage in specialized services or project-based work such as consulting, graphic design, or skilled trades for multiple clients. In this sense, the gig economy reflects a digitally mediated subset of a broader and increasingly common set of employment structures that fall outside the traditional full-time model.

To reflect this broader scope, this study adopts the term alternative work arrangements to encompass a wide range of nonstandard employment relationships, consistent with the Bureau of Labor Statistics’ definition. As of July 2023, approximately 11.9 million people identified as

independent contractors in their main job, representing 7.4% of total U.S. employment, up from 6.9% in May 2017, highlighting the growing relevance of nontraditional work arrangements (BLS, 2024). More broadly, alternative work arrangements, including independent contractors, on-call workers, contract firm employees, and temporary help agency workers, now comprise nearly 10% of the U.S. workforce (BLS, 2024). Drawing data from the General Social Survey (GSS), which includes respondent self-reports of employment status, this research empirically examines how alternative work arrangements relate to workers' subjective well-being (SWB).

Subjective well-being refers to individuals' self-assessed happiness, life satisfaction, and emotional experiences. In a work context, SWB is shaped by more than just wages; it is influenced by factors like job autonomy, income stability, perceived fairness, interpersonal relationships, and supportive work environments (Shier & Graham, 2010). While some individuals pursue alternative work arrangements to gain greater flexibility or independence, others may enter them out of economic necessity or limited options. These differing motivations and entry points likely translate into very different work experiences and thus SWB outcomes.

Much of the existing literature on gig work has focused on low-wage, app-based service work. This thesis broadens the scope by examining how subjective well-being varies across worker subgroups in AWAs, with particular attention to differences in managerial and professional status. This approach sheds light on whether individuals in supervisory or professional roles experience AWAs differently from those in non-supervisory or lower-skilled positions. While higher-status workers may enjoy greater autonomy and earning potential, they remain susceptible to challenges such as income volatility, limited institutional support, and restricted access to benefits. Examining their well-being can offer insights into labor policies, worker protections, and career planning in an evolving labor market.

To analyze the relationship between employment type and SWB, this study applies a multi-stage regression strategy. First, it examines how happiness and job satisfaction vary between regular and alternative workers. Then, it investigates whether job satisfaction may act as a mediator, potentially explaining part of the relationship between work arrangement and overall happiness. Since job satisfaction reflects individuals' evaluation of their work environment, a major life domain, it is a theoretically meaningful channel through which employment structure could affect broader life satisfaction. My findings show that well-being outcomes vary across alternative work arrangements. Independent contractors report higher job satisfaction and modestly greater happiness than regular employees. In contrast, on-call and contract firm workers exhibit lower or comparable well-being, with on-call workers reporting significantly lower job satisfaction. Mediation analysis confirms that job satisfaction explains much of the happiness advantage among contractors. These results highlight the importance of distinguishing between AWAs types and suggest autonomy is a key driver of subjective well-being.

2. Related Literature

Substantial research has examined the motivations for participating in nonstandard forms of work, particularly within the gig economy. Scholars commonly distinguish between two types of gig work based on task structure and delivery: crowd work and work-on-demand (Klein et al., 2024). Crowd-work typically involves remote, digitally mediated services such as graphic design, while work-on-demand refers to physical, task-based roles such as ridesharing or food delivery, requiring minimal specialized training (Teodoro et al., 2014).

Across contexts, motivations for engaging in AWAs commonly center around autonomy, flexibility, and opportunities to generate income outside traditional full-time employment. For

some, especially those in specialized or project-based roles, alternative arrangements offer greater control over work schedules, choice of clients, and the ability to build a diverse portfolio (Roy et al., 2020; Spreitzer et al., 2017). These features can foster a sense of independence and fulfillment that is not always present in traditional employment structures. For others, particularly those in low-wage or on-demand service roles, participation may be driven more by financial necessity or limited access to stable, full-time employment (Teodoro et al., 2014). These contrasting motivations imply a high degree of heterogeneity within the AWAs population, which in turn may shape outcomes related to well-being.

This heterogeneity appears in studies examining subjective well-being (SWB) among alternative workers. Van der Zwan, Hessels, and Burger (2019), for example, drew data from the UK Household Longitudinal Study (UKHLS), using survey waves from 2009 to 2015. Employing both cross-sectional ordinary least squares (OLS) and longitudinal fixed-effects analyses, they found that freelancers report higher satisfaction with both work and leisure, suggesting that the flexibility and autonomy associated with freelance work may enhance wellbeing despite financial uncertainty.

Conversely, Wang, Li, and Coutts (2022) analyzed cross-sectional data from UKHLS Wave 11 (2019–2021). Using weighted OLS regression, they found that gig workers reported lower life satisfaction and worse mental health outcomes compared to traditional employees, although they fared better than the unemployed. The authors highlighted financial precarity and social isolation as key mediators of these negative outcomes.

The discrepancy in findings between these studies stems primarily from methodological sample differences. Zwan's study focuses exclusively on freelancers, which are often assumed to be higher-skilled and more autonomous, while Wang's study adopts a broader definition of gig

work that includes both skilled and lower-skilled on-demand workers. Since financial instability and social isolation are more common among low-skilled gig workers, this broader inclusion may contribute to Wang et al.'s more negative findings. Therefore, when researching nontraditional work arrangements, it is crucial to clearly delineate the specific subset of individuals being analyzed. Without accounting for this heterogeneity, comparisons across studies can yield conflicting conclusions.

This study contributes to existing literature by formally examining its mediating role between employment structure and broader life satisfaction. Since job satisfaction captures workers' day-to-day evaluations of their work experiences, it likely influences how workers feel about their lives overall. This thesis contributes to the literature by addressing two important gaps. First, it adopts a comprehensive view of alternative work arrangements, encompassing not only freelancers and gig workers, but also on-call employees and contract firm workers. Second, it formally tests the extent to which job satisfaction mediates the relationship between alternative work arrangements and overall happiness, providing new insights into how different work structures affect subjective well-being.

3. Data and Variables

Dataset

The dataset utilized in this study is derived from the General Social Survey (GSS), a publicly available dataset that collects extensive self-reported information on the characteristics, attitudes, and behaviors of the American population (NORC at the University of Chicago, 2021). Central to this analysis is the GSS question that classifies respondents into categories such as independent contractor, temp-agency worker, on-call worker, contract-company worker, and regular employee: "How would you describe your work arrangement in your main job?" These

classifications serve as the primary identifier for alternative worker status, with data extracted from survey waves 2006, 2010, 2014, 2018, and 2022, the waves in which this question was asked (Appendix Table A2). Although the GSS is a long-running national survey, it does not track individuals over time due to anonymity protections, making longitudinal analysis infeasible. As a result, this study employs a cross-sectional design, comparing different worker groups at specific time points rather than tracking individual changes.

In addition to employment type, the dataset includes a range of variables related to wellbeing and job quality. The primary outcomes of interest are happiness and job satisfaction, each based on single-item survey questions with ordinal response categories (full question wording provided in Appendix Table A1).¹ Although these responses are ordinal in nature, I treat them as continuous variables in the regression analysis to facilitate interpretation and allow for linear estimation of average effects. Additional covariates include demographic characteristics (education, marital status, region) and work characteristics (industry, managerial occupation, occupational status, weeks worked). Following Budig (2006) and Carleton and Kelton (2018), I use the 2010 Census Occupation Codes to identify and control for individuals classified as managerial or professional.² These factors provide a comprehensive basis for analyzing differences in work satisfaction across employment types.

¹ Responses include "Not too happy," "Pretty happy," and "Very happy" for happiness; and "Very dissatisfied", "A little dissatisfied," "Moderately satisfied," and "Very satisfied" for job satisfaction.

² Individuals classified under the 2010 Census occupation codes 10–3540 are labeled "Managerial, Professional, and Related Occupations." These jobs typically require a college education, involve high-level cognitive and technical skills, and often require formal credentials or licenses.

Summary Statistics

Table 1 provides descriptive statistics for key demographic, socioeconomic, and subjective well-being variables, broken down by work arrangement type, based on a sample of approximately 7,800 respondents with complete data (Appendix Tables A1 and A2 for variable definitions). Respondents in alternative work arrangements are significantly more likely than regular workers to report being "very happy" (32% vs. 28%) and "very satisfied" with their jobs (55% vs. 47%). They typically experience lower average income (\$29,524 vs. \$33,912), shorter annual workweeks (42.02 vs. 47.21 weeks), and lower educational attainment (13.78 vs. 14.31 years). In terms of occupational and demographic characteristics, alternative workers are less likely to be employed in managerial or professional roles (33% vs. 41%), are more likely to be male (59% vs. 46%), and are slightly older on average (45.92 vs. 42.30 years). The two groups do not differ significantly in marital status.

Appendix Figure A1 visualizes trends in job satisfaction and happiness from 2006 to 2022, showing that alternative workers consistently report higher levels of job satisfaction, while happiness levels remain relatively stable and comparable between the two groups.

Together, these descriptive differences provide preliminary support for the flexibility hypothesis, suggesting that the non-monetary benefits associated with alternative work, such as autonomy or improved work-life balance, may offset certain economic drawbacks. Together, these patterns offer important context for the multivariate analyses that follow.

Table 1: Summary Statistics by Work Arrangement Status

	Overall N = 6,892	Regular Worker N = 5,536	Alternative Worker N = 1,356
General Happiness			
Not too happy	13%	12%	16%***
Pretty happy	58%	60%	53%***
Very happy	29%	28%	32%***
Job Satisfaction			
Very dissatisfied	4%	4%	3%
A little dissatisfied	9%	9%	7%**
Moderately satisfied	39%	40%	35%***
Very satisfied	48%	47%	55%***
Professional Occupation	40%	41%	33%***
Education (years)	14.21 (2.83)	14.31 (2.79)	13.78 (2.96)***
Income (constant \$)	33,049 (34,740)	33,912 (33,276)	29,524 (39,988)***
Age	43.01 (13.55)	42.30 (13.19)	45.92 (14.55)***
Gender %			
MALE	48%	46%	59%***
FEMALE	52%	54%	41%***
Weeks Worked Last Year	46.19 (11.94)	47.21 (10.88)	42.02 (14.82)***
Married	46%	47%	45%

Note: Means and proportions are reported by work arrangement type (regular vs. alternative). Statistical significance indicates differences relative to regular workers, based on two-sample t-tests (for continuous variables) and two-sample proportion tests (for categorical variables).

*** p<0.01, ** p<0.05, * p<0.1

Table 2 shows the distribution of employment types from 2002 to 2022, revealing that regular permanent employment consistently comprised approximately 80% of the workforce each year. The proportion of alternative workers remained relatively stable at around 20%, despite fluctuations in their absolute numbers over time (Appendix Figure A2).

Within the alternative worker category, independent contractors represent the largest and most stable subgroup, consistently comprising over 12% of the total workforce. In contrast, on-call workers and those employed by contractors exhibit greater year-to-year variability, suggesting increased sensitivity to labor market dynamics or sampling variation (Appendix Figure A3). Given the consistently small sample size of respondents “paid by a temporary agency” (less than 2% in all years), this category will be excluded from all future analyses.

Table 2: Work Type Distribution by Year

Work Type	2006	2010	2014	2018	2022
Regular, permanent employee	79.86%	80.06%	79.9%	79.97%	81.53%
Independent contractors	12.75%	12.37%	13.77%	12.21%	12.61%
Work for contractors who provide services	3.63%	2.86%	2.91%	3.65%	3.29%
On-call, work only when called to work	2.59%	3.51%	2.82%	2.68%	1.85%
Paid by a temporary agency	1.17%	1.2%	0.6%	1.49%	0.72%
Count	1624	1083	1169	1343	1673

Appendix Table A3 presents summary statistics by worker type, excluding those paid by temporary agencies. Among the groups, on-call workers appear to be the most economically disadvantaged, with the lowest average income (\$14,588), lowest educational attainment (12.77 years), and fewest weeks worked annually (34.02). These patterns reflect the instability of oncall roles and may help explain their relatively high levels of job dissatisfaction—14% report being “a little dissatisfied,” compared to just 9.3% of regular employees.

By contrast, independent contractors report the highest job satisfaction, with 64% “very satisfied,” and the highest share of “very happy” respondents (35%). Although their income is similar to regular employees, they appear to benefit more from the autonomy and flexibility typically associated with self-directed work. However, these advantages are not shared across all alternative arrangements. On-call and contract workers report lower well-being and are less likely to hold professional roles, suggesting that the benefits of non-traditional work depend on the degree of control and stability the role provides.

Demographically, independent contractors skew older (47.8 years) and are more male (60%), while regular employees are more gender-balanced (54% female). On-call and contract workers are also less likely to be in higher-level managerial or professional roles.

The findings highlight structural disparities in work experience. On-call workers face the greatest economic precarity, while independent contractors enjoy more satisfaction despite

similar income levels. Work arrangements thus appear closely tied to both economic security and subjective well-being.

Appendix Table A4 compares industry employment characteristics by Work Arrangement using data recategorized by the 2022 North American Industry Classification System (NAICS). It presents total employment, the number and share of regular employees versus alternative workers, and average job satisfaction on a 1–4 scale. The share of alternative workers within industries varies considerably. The highest proportions are found in Agriculture, Forestry, and Fishing (50.7%), Construction (47.0%), and Real Estate and Leasing (45.2%). Although alternative arrangements are often linked to flexibility and independence, their concentration in labor-intensive sectors points to the growing role of nonstandard work in physically demanding and project-based industries.

In contrast, industries like Public Administration (6.1%), Educational Services (9.1%), and Accommodation and Food Services (9.4%) show much lower alternative worker shares. These differences likely reflect more rigid employment structures, stronger regulatory oversight, and a greater presence of long-term contracts or unionized labor (BLS 2025). Interestingly, Management of Companies and Enterprises sector reports no alternative workers and few regular employees, though this may be due to data limitations.

Job satisfaction scores are generally consistent across work arrangements, with only minor differences between regular and alternative workers within most industries. Overall, the findings highlight the diverse presence of alternative work across the labor market. They appear not only in flexible or entrepreneurial fields but also in sectors reliant on seasonal or contract-based labor, suggesting that alternative arrangements are reshaping work across both high- and

low-skill domains, with important implications for employment stability, worker protections, and policy design.

4. Empirical Strategy

This thesis investigates whether alternative work arrangements are associated with differences in subjective well-being, and whether this relationship operates through the channel of job satisfaction. The analysis proceeds in three main steps:

Step 1: Baseline Regressions

I begin by estimating OLS regressions to assess how employment type relates to two outcomes: general happiness and job satisfaction. Both models control for a consistent set of covariates – professional status, education, income, age, marital status, sex, and race – summarized as X_{it} . Additionally, fixed effects for region γ_r , industry δ_j , and survey year λ_t are included to account for differences in labor markets, sectoral norms, and macroeconomic conditions.

The baseline models are:

$$Happy_{it} = \beta_0 + \beta_1 AlternativeWorker_{it} + \beta_2 X_{it} + \gamma_r + \delta_j + \lambda_t + \epsilon_{it}$$

$$JobSat_{it} = \beta_0 + \beta_1 AlternativeWorker_{it} + \beta_2 X_{it} + \gamma_r + \delta_j + \lambda_t + \epsilon_{it}$$

X_{it} captures effects for professional status, education, income, age, marital status, sex, and race

Step 2: Mediation via Job Satisfaction

Next, I test whether job satisfaction mediates the relationship between employment type and happiness by adding job satisfaction as a covariate in the happiness model:

$$Happy_{it} = \beta_0 + \beta_1 AlternativeWorker_{it} + \beta_2 JobSat_{it} + \beta_3 X_{it} + \gamma_r + \delta_j + \lambda_t + \epsilon_{it}$$

If the coefficient for alternative worker status declines notably when job satisfaction is added, it suggests that job satisfaction may be a mediating mechanism linking employment type to overall happiness.

Step 3: Formal Mediation Analysis

Finally, I conduct a formal mediation analysis to quantify the direct and indirect effects of employment type on happiness, using the standard two-equation framework. This approach tests whether the effect of employment arrangement on happiness operates through job satisfaction, a theoretically justified mediator given its close link to daily work experiences and overall well-being.

$$JobSat_{it} = \alpha_0 + \alpha_1 AlternativeWorker_{it} + \alpha_2 X_{it} + \gamma_r + \delta_j + \lambda_t + \epsilon_{it}$$

$$Happy_{it} = \beta_0 + \beta'_1 AlternativeWorker_{it} + \beta_2 JobSat_{it} + \beta_3 X_{it} + \gamma_r + \delta_j + \lambda_t + \epsilon_{it}$$

Here, β'_1 captures the direct effect of employment type on happiness, while the indirect effect operates through job satisfaction.

5. Results

5.1 Baseline Associations: Happiness and Job Satisfaction

Table 5 reports baseline OLS regression results for general happiness (Panel A) and job satisfaction (Panel B), using regular employees as the reference group. Each panel includes both an unadjusted model and a fully specified model with demographic controls and fixed effects.

General Happiness (1–3 scale):

In the unadjusted model, regular employees report an average score of 2.156, slightly above "Pretty happy." After controlling for covariates, independent contractors show a significant happiness advantage ($\beta = 0.072$, $p < 0.01$; see column 4). On-call and contract workers do not differ significantly from regular employees, and their negative coefficients diminish once controls are added, suggesting these gaps are attributable to factors like income, education, and marital status. Among covariates, professional status, education, income, and marital status are positively associated with happiness, while age is negatively associated. Women report slightly higher levels of happiness than men, and race is not a significant predictor.

Job Satisfaction (1–4 scale):

In the unadjusted model, regular employees average 3.306, just above "Moderately satisfied." Independent contractors report significantly higher satisfaction ($\beta = 0.217$, $p < 0.01$; see column 8), reinforcing the value of autonomy in shaping work experiences. On-call workers report significantly lower satisfaction ($\beta = -0.124$, $p < 0.05$, see column 8), even after full adjustment, indicating that their dissatisfaction is not fully explained by demographic or

socioeconomic factors. Contract workers do not differ significantly from regular employees in their jobs, likely due to structural similarities in job roles or workplace environments. Among covariates, while age is negatively associated with happiness, it is positively associated with job satisfaction, suggesting that older workers may adapt more positively to their work environment over time. Black respondents report significantly lower job satisfaction in the fully adjusted models, raising potential concerns about racial disparities in workplace experience.

Taken together, the results suggest that independent contractors benefit from both higher job satisfaction and overall happiness, while on-call workers' dissatisfaction is more limited to their work experience and does not translate into broader reductions in well-being. For contract-firm employees, no significant differences are observed in either job satisfaction or happiness, indicating a more neutral relationship with subjective well-being.

Fixed Effects Interpretation:

Appendix Table A5 provides estimates from the fully specified models in Table 5, offering a surface-level view of how region, year, and industry shape subjective well-being. Regional fixed effects show that respondents in the Mountain and West South-Central regions report higher happiness ($\beta = 0.053$, $p < 0.1$; $\beta = 0.062$, $p < 0.05$, respectively), with the West South-Central also linked to higher job satisfaction ($\beta = 0.122$, $p < 0.01$). These differences may reflect regional variation in labor markets or the cost of living.

Year effects, using 2006 as the reference, reveal that happiness declined in 2010 ($\beta = -0.076$, $p < 0.01$) and more sharply in 2022 ($\beta = -0.147$, $p < 0.01$), while job satisfaction also fell modestly in 2022 ($\beta = -0.047$, $p < 0.1$), possibly reflecting the lasting impact of the Great Recession and COVID-19.

Industry fixed effects show that workers in utilities, construction, health care, education services, and public administration report significantly higher job satisfaction than those in accommodation and food services (reference). These differences likely reflect greater stability, clearer career paths, or more meaningful work. In contrast, such patterns are less evident for general happiness as fewer coefficients reach significance. This likely means that job satisfaction is more directly influenced by workplace characteristics, while happiness may depend more on broader environmental factors.

Table 5: OLS: Happiness and Job Satisfaction by Work Type

	Panel A: General Happiness				Panel B: Job Satisfaction			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Employment Relationships								
Independent contractors	0.063*** (0.024)	0.079*** (0.024)	0.076*** (0.024)	0.072*** (0.025)	0.251*** (0.025)	0.226*** (0.025)	0.225*** (0.025)	0.217*** (0.027)
On-call, work only when called	-0.061 (0.048)	-0.000 (0.046)	-0.007 (0.047)	-0.010 (0.047)	-0.145** (0.059)	-0.102* (0.057)	-0.101* (0.058)	-0.124** (0.058)
Work for contractors who provide workers/services	-0.072 (0.046)	-0.039 (0.044)	-0.042 (0.044)	-0.041 (0.045)	-0.008 (0.050)	0.025 (0.049)	0.021 (0.050)	-0.016 (0.050)
Covariates								
Managerial/Professional		0.033* (0.018)	0.037** (0.018)	0.035* (0.019)		0.125*** (0.022)	0.126*** (0.022)	0.095*** (0.023)
Education (Years)		0.005* (0.003)	0.007** (0.003)	0.007** (0.003)		0.002 (0.004)	0.004 (0.004)	0.001 (0.004)
Income (thousands)		0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)		0.001*** (0.000)	0.001*** (0.000)	0.001*** (0.000)
Age		-0.002*** (0.001)	-0.002*** (0.001)	-0.002*** (0.001)		0.006*** (0.001)	0.006*** (0.001)	0.006*** (0.001)
Married		0.308*** (0.015)	0.304*** (0.015)	0.303*** (0.015)		0.104*** (0.019)	0.100*** (0.019)	0.095*** (0.019)
Female		0.044*** (0.015)	0.043*** (0.015)	0.038** (0.016)		0.033* (0.019)	0.031 (0.019)	0.014 (0.021)
Black		0.005 (0.022)	0.000 (0.022)	0.001 (0.022)		-0.039 (0.028)	-0.052* (0.028)	-0.054* (0.028)
Other Race		0.003 (0.025)	0.002 (0.025)	-0.000 (0.025)		0.030 (0.029)	0.031 (0.030)	0.034 (0.030)
Constant	2.156*** (0.008)	1.931*** (0.050)	1.938*** (0.054)	1.941*** (0.061)	3.306*** (0.011)	2.870*** (0.061)	2.823*** (0.067)	2.794*** (0.076)
Fixed Effects								
Survey FE	No	No	Yes	Yes	No	No	Yes	Yes
Region FE	No	No	Yes	Yes	No	No	Yes	Yes

Year FE	No	No	Yes	Yes	No	No	Yes	Yes
Industry FE	No	No	No	Yes	No	No	No	Yes
Observations	6,821	6,821	6,821	6,821	6,821	6,821	6,821	6,821

Note: Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

5.2 Mediation through Job Satisfaction

To explore whether job satisfaction explains part of the observed happiness advantage among independent contractors, **Table 6** introduces job satisfaction as a covariate in models predicting general happiness.

In Column 1, consistent with Table 5, independent contractors report a significant happiness advantage ($\beta = 0.063$, $p < 0.01$). However, after controlling job satisfaction in Model 2, the coefficient drops to 0.011 and loses statistical significance, suggesting that job satisfaction accounts for much of the observed difference. Columns 3 and 4 sequentially add demographic and fixed effects. Although the coefficient for independent contractors rises slightly ($\beta = 0.031$ in Model 4), it remains statistically insignificant.

Across all specifications, job satisfaction emerges as a strong and consistent predictor of happiness ($\beta = 0.114$, $p < 0.01$), underscoring its central role in connecting work structure to broader life satisfaction. Meanwhile, the coefficients for on-call and contract workers remain negative and insignificant throughout, consistent with earlier findings that these groups do not enjoy happiness advantages.

Table 6: Extended Happiness model with Job Satisfaction Covariate

	General Happiness (1)	General Happiness (2)	General Happiness (3)	General Happiness (4)
Employment Relationships				
Independent contractors	0.063*** (0.024)	0.011 (0.023)	0.036* (0.023)	0.031 (0.024)
On-call worker	-0.061 (0.048)	-0.031 (0.048)	0.019 (0.047)	0.013 (0.047)
Work for contractors who provides services	-0.072 (0.046)	-0.071 (0.044)	-0.044 (0.042)	-0.038 (0.043)
Job Satisfaction		0.209*** (0.010)	0.189*** (0.010)	0.1186*** (0.010)
Covariates				
Managerial/Professional			0.009 (0.017)	0.018 (0.018)
Education (years)			0.005 (0.003)	0.007** (0.003)
Income (thousands)			0.001*** (0.000)	0.001*** (0.000)
Age			-0.003*** (0.001)	-0.003*** (0.001)
Married			0.289*** (0.015)	0.285*** (0.015)
Female			0.038** (0.015)	0.036** (0.016)
Black			0.012 (0.021)	0.011 (0.022)
Other Race			-0.003 (0.024)	-0.006 (0.024)
Constant	2.156*** (0.008)	1.465*** (0.034)	1.389*** (0.057)	1.420*** (0.066)
Fixed Effects				
Survey FE	No	No	No	Yes
Region FE	No	No	No	Yes
Industry FE	No	No	No	Yes
Year FE	No	No	No	Yes
Observations	6,821	6,821	6,821	6,821

Note: Robust standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

5.3 Mediation Analysis

Mediation results presented in **Table 7** demonstrate that job satisfaction plays a key explanatory role in the happiness advantage observed among independent contractors. After adjusting for covariates, about 53.2% of the total effect of being an independent contractor on happiness is mediated by job satisfaction. This suggests that much of their well-being advantage stems from greater day-to-day work satisfaction. These findings align with earlier results in Table 6, where simply adding job satisfaction to the model (Column 2) causes the coefficient for independent contractor status to drop sharply from 0.063*** to 0.011 and lose statistical significance. This pattern indicates that independent contractors are happier largely because they are more satisfied with their jobs, pointing to work autonomy and better alignment with personal preferences as likely sources of higher well-being.

Interestingly, on-call workers exhibit a negative indirect effect of job satisfaction on happiness (−0.0254) despite a small, non-significant total effect (−0.0141), yielding a percent mediated value over 180%. This reflects a suppression effect, where the indirect and direct effects move in opposite directions (MacKinnon et al, 2000). In essence, on-call workers' lower job satisfaction masks a small, positive direct association between on-call work and happiness.³ In practical terms, this suggests that while on-call workers may be dissatisfied with their jobs, due to factors such as unstable hours, limited benefits, and scheduling unpredictability, certain

³ For on-call workers, the total effect on happiness (−0.061; Table 6, Column 1) includes not only the direct impact of being on-call, but also an indirect pathway:

On-call status → lower job satisfaction → lower happiness

Since job satisfaction is positively associated with happiness, this negative indirect path makes the total effect appear more negative than it truly is. When job satisfaction is excluded, the indirect pathway is absorbed into the total estimate, exaggerating the harmful effect of on-call work. Once job satisfaction is included (Column 2), the coefficient shrinks to −0.031, revealing that the original estimate overstated the true relationship. This pattern reflects a suppression effect: lower job satisfaction among on-call workers masked a more neutral direct association with happiness.

features like flexibility, autonomy, or minimal oversight may still contribute positively to their well-being.

For contract workers, the mediated effect is modest (5.2%) and not statistically significant, suggesting job satisfaction has little influence on their happiness. This is further supported by the minimal change in the coefficient for “work for contractors” (from -0.072 to -0.071 ; Table 6, Columns 1–2) after controlling for job satisfaction, indicating no meaningful mediation or suppression. These findings show that job satisfaction plays different roles across work types: a mediator for independent contractors, a suppressor for on-call workers, and largely irrelevant for contract-firm employees.

This reinforces a key point from earlier analyses: alternative work arrangements are not experienced equally. Independent contractors appear to benefit through higher job satisfaction, likely due to greater autonomy and a better fit with personal preferences. For other nonstandard workers, however, job satisfaction does not explain variation in well-being, underscoring the need to look beyond averages and examine the specific mechanisms shaping subjective outcomes.

Table 7: Mediation Analysis

Employment Relationships (ref.= Regular Employee)	Indirect Effect	Direct Effect	Total Effect	Perfect Mediated
Panel A: Happiness				
Independent contractors	0.0413*** (0.0056)	0.0363* (0.0216)	0.0776* (0.0221)	53.2%
On-call, work only when called	-0.0254** (0.0110)	0.0113 (0.0445)	-0.0141 (0.0458)	180.1%
Work for contractors who provides services	-0.0028 (0.0097)	-0.0508 (0.3955)	-0.0536 (0.0407)	5.2%

Note: Robust standard errors in parentheses, *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

Mediated Direct Effects are derived from SEM using Job Satisfaction as the mediator. Percent Mediated is calculated as indirect effects divided by total effects, where total effect is the sum of indirect and direct effect

6. Robustness

To ensure reliability of the main results, several robust checks were conducted. These fall into three broad categories: accounting for within-group correlation, testing alternative model specifications, and exploring heterogeneity across subgroups and over time. While job satisfaction is shown to mediate the relationship between employment type and happiness, I do not include it as a covariate in robustness checks. Including job satisfaction would likely introduce post-treatment bias, as it is influenced by employment type and helps explain how employment type affects happiness. To maintain a clear interpretation of the total effect of work arrangement on subjective well-being, job satisfaction is excluded from these models.

Across all checks, the core result remains unchanged: independent contractors report higher job satisfaction and happiness, on-call workers show persistently lower job satisfaction, and in some models, significantly lower happiness; and contract firm workers exhibit no statistically meaningful differences in well-being compared to regular employees.

6.1 Adjusting for Grouped Error Structures and Sector-Specific Bias

The baseline regressions assume that individual workers' error terms are independent across observations. However, workers within the same industry may be jointly exposed to industry-specific shocks, such as regulatory changes in healthcare or sectoral downturns in manufacturing, that influence both happiness and job satisfaction. To account for shared industry effects, all models were re-estimated, clustering standard errors at the industry level. Positive associations between independent contracting with happiness and job satisfaction remain statistically significant, while on-call work continues to show a significant negative relationship

with job satisfaction. These findings indicate that the main results are not driven by unobserved industry-specific factors affecting the worker's well-being (Appendix Table A6)

I also tested for within-industry heterogeneity by estimating the relationship between employment type and both happiness and job satisfaction for each sector. Statistically significant effects of employment type are more consistently observed for job satisfaction than for happiness (Appendix Table A7). Independent contractors report significantly higher job satisfaction in most sectors (9 out of 16), reinforcing the value of autonomy and flexibility in the workplace. In contrast, on-call workers frequently report lower job satisfaction, particularly in accommodation and food services, construction, and finance.

Significant differences in happiness are less common and show no consistent pattern, suggesting that broader life satisfaction may depend on factors beyond the work context. One exception is the lower happiness reported by on-call workers in professional services and public administration. Overall, these findings underscore the importance of accounting for industry context when evaluating the effects of alternative work arrangements on worker well-being, as sectoral differences appear to exert a stronger influence on job satisfaction than on broader measures of happiness.

6.2 Alternative Estimation Methods and Functional Forms

In this analysis, happiness and job satisfaction were treated as approximately continuous variables and estimated using OLS. However, both measures are ordinal, with respondents selecting from ordered but discrete categories. As a result, the OLS assumption of equal spacing between categories may not fully hold; for example, the jump from “Not too happy” to “Pretty happy” may not carry the same weight as the jump from “Pretty happy” to “Very happy.”

To address this, Appendix Table A8 presents results from ordered logistic regression models to assess the robustness of our findings to this measurement issue. The ordered logit findings remain consistent with the OLS estimates: independent contractors are significantly more likely to report higher well-being, with a 0.258 increase in the ordered logit coefficient for happiness and a 0.634 increase in job satisfaction (both significant at the 1% level). On-call workers show significantly lower job satisfaction (around -0.343 , $p\text{-value} < 0.01$), but no significant difference in happiness. Contract workers exhibit small, negative associations with both outcomes, but again, not significant.

To explore potential nonlinear relationships, I also included quadratic terms for both age and income in Appendix Table A8. Prior research suggests that the association between age and happiness often follows a U-shaped trajectory, in which well-being declines during midlife and rebounds in older age (Blanchflower et al., 2008). However, the results here do not strongly support that pattern. In the happiness models, the coefficient for age-squared is small and not statistically significant, indicating no meaningful curvature. Interaction terms between employment type and age-squared are also insignificant, suggesting that the relationship between age and happiness remains largely linear across work arrangements.

The findings are similar for job satisfaction. Most age-squared terms and their employment-type interactions are not statistically significant. A modest exception appears among on-call workers, where the interaction with age-squared is small but statistically significant ($p < 0.10$), suggesting a weak nonlinear relationship. Still, the linear age coefficient is negative, indicating that job satisfaction tends to decline with age, though the magnitude of this effect is relatively modest.

Income, by contrast, exhibits a statistically significant and positive association with both happiness and job satisfaction in the ordered logit models (Columns 1–2). This relationship remains robust even after including quadratic income terms (Column 3). Both the quadratic terms and their interaction effects with worker types are generally insignificant, offering limited evidence of diminishing returns. The notable exception is again among on-call workers. For this group, income has a strong positive effect on job satisfaction, while the squared income term is negative and statistically significant ($p < 0.01$). This pattern suggests that although job satisfaction rises with income for on-call workers, the gains diminish at higher income levels. Overall, the results suggest that income is positively and linearly associated with happiness across the broader workforce, while its relationship with job satisfaction is more limited and primarily concentrated among on-call workers.

6.3 Temporal Stability and Demographic Subgroup Effects

When incorporating year fixed effects, the analysis controls for time-specific shocks, such as recessions or pandemics, that uniformly influence subjective well-being across the labor force. However, this approach implicitly assumes that the effect of being in an alternative work arrangement is stable across time. To assess whether this assumption holds, interaction terms between employment type and survey year were included in fully specified models, allowing the effect of alternative employment to vary over time.

Appendix Table A10 presents fully adjusted regression results with clustered standard errors. The relationship between employment type and well-being remains broadly consistent with earlier models. Independent contractors report significantly higher happiness and substantially higher job satisfaction compared to regular workers. On-call workers show

significantly lower happiness, though their job satisfaction is not statistically different. Notably, this represents a shift from earlier results (Table 5), where only job satisfaction was significant for on-call workers. Contract workers do not differ significantly from regular employees on either outcome.

Temporal variation is most evident among regular workers. Happiness declined sharply in 2010 and 2022 ($p < 0.001$), reflecting the impacts of the Great Recession and the COVID-19 pandemic. In contrast, job satisfaction rose modestly but significantly in 2014 ($p < 0.05$), coinciding with a labor market recovery driven by gains in higher-paying sectors such as professional and business services, the first such shift since 2008. This has likely contributed to improved morale among permanent employees. In contrast, no clear time trends emerge for workers in alternative arrangements. Their predicted happiness and job satisfaction remain relatively stable over the years.

Appendix Figures A4 and A5 plot predicted values from Appendix Table A11 that are derived by summing the relevant coefficients from the regression results in Appendix Table A10. While most interaction effects lack statistical significance, the figures illustrate general patterns: regular workers' happiness appears most sensitive to economic shocks, independent contractors maintain consistently high satisfaction, and other alternative workers show variability without clear trends. These visualizations are descriptively informative but should be interpreted with caution due to limited statistical support.

Lastly, I tested for heterogeneous effects by interacting employment type with gender, marital status, and professional occupational status. While women, married individuals, and professionals report higher well-being overall, the interaction terms are not statistically

significant (Appendix Table A11). This indicates that the effects of alternative work arrangements on subjective well-being are broadly similar across demographic subgroups.

7. Discussion

Key Findings

The central empirical insight is that independent contractors exhibit higher happiness and satisfaction compared to regular employees. This advantage is largely mediated by job satisfaction, suggesting that the psychological benefits of autonomy, self-direction, and work alignment with personal values play a central role in shaping life satisfaction.

On the other hand, on-call workers, often identified as precarious labor due to job insecurity, unstable schedules, and limited benefits (Menéndez et al., 2007), report significantly lower job satisfaction than regular employees. Notably, they show no happiness advantage once job satisfaction is controlled. This may indicate a suppression effect, whereby dissatisfaction with the structural features of on-call work (e.g., unstable hours and lack of control) counteracts any potential upsides, such as flexible or reduced work demands.

Contract firm workers, those employed by third-party staffing agencies but working full-time alongside regular staff, show no significant differences from regular employees in either job satisfaction or happiness. This finding is consistent with existing research suggesting that such workers often perform similar tasks, effectively replicating the experience of standard employment, without necessarily being offered the autonomy associated with independent contracting or the instability of on-call work (Houseman et al., 2003).

Importantly, these results are consistent across various model specifications, including ordered logistic regressions and robust checks accounting for sectoral, temporal, and

demographic variation. Theoretically, these findings challenge simplistic narratives that either criticize alternative work for its precarity or romanticize nonstandard work for its flexibility. Instead, my findings highlight the importance of autonomy and voluntariness as a mediating factor in shaping workers' well-being. The higher satisfaction reported by independent contractors may reflect both actual job characteristics and a self-selection effect, where individuals who value independence and flexibility may be more likely to enter and perform well in these roles.

This interpretation aligns with self-determination theory, which emphasizes autonomy as a fundamental element of human motivation and well-being (Ryan & Deci, 2000). In contrast, the dissatisfaction among on-call workers is likely rooted in externally imposed constraints such as irregular schedules, uncertain hours, and lack of benefits. These are all conditions that erode perceived control and predictability, both of which are detrimental to well-being.

Policy Implications

These findings have important consequences for labor market regulation and social protection policy. They show that a one-size-fits-all approach to regulating nonstandard work simply does not work. Although often grouped together under the same umbrella, independent contractors, on-call workers, and contract firm employees lead very different work lives. For policy to be effective, it must adopt a differentiated approach tailored to the needs of each arrangement.

Although independent contractors report greater well-being, they lack access to many basic protections provided to standard employees. Independent contractors are excluded from wage and hour laws, collective bargaining rights, unemployment insurance, and employer-

sponsored benefits such as health insurance and retirement plans (Abraham et al., 2023). To support this group without compromising their valued independence, policies should prioritize portable benefits systems, streamlined tax filing tools for the self-employed, and universal access to health insurance and retirement accounts.

By contrast, on-call workers demonstrate lower job satisfaction even after accounting for income and demographics. This suggests that dissatisfaction stems from the structural nature of the work arrangements themselves. Irregular and unpredictable hours, limited control over scheduling, and a lack of access to benefits can create a sense of instability, thus undermining well-being. Previous research has linked on-call and irregular shifts to higher levels of work-family conflict and stress (Golden & Kim, 2017). Addressing these issues will require targeted policy solutions such as fair scheduling legislation, minimum guaranteed hours, and expanded access to unemployment and paid leave. These reforms can mitigate the instability of on-call work.

Contract firm employees, meanwhile, report well-being levels like regular employees. However, their employment status places them in a legal gray zone, where they may be excluded from protections afforded to direct hires. Ensuring equal treatment, such as safety standards and access to benefits, would help prevent the erosion of labor standards in third-party employment structure.

As alternative work arrangements continue to comprise a substantial portion of the modern labor force, it is essential to continually adapt regulatory frameworks to ensure that these workers are not excluded from critical protections such as health insurance, retirement security, and stable income sources. Evidence also suggests that alternative work is more prevalent among marginalized communities, particularly low-income individuals and those at risk of welfare

dependency (Lane et al., 2003). Hence, upholding equitable labor standards not only protects individual workers and addresses broader labor market inequality, but also promotes inclusive economic growth and safeguards long-term economic stability and productivity.

Limitations

Several limitations should be noted. First, the use of cross-sectional data limits the ability to draw causal conclusions. As mentioned previously, it is plausible that individuals with higher baseline happiness or a greater preference for autonomy are more likely to self-select into independent contracting, rather than the work arrangement itself causing greater well-being. Without longitudinal data that tracks individuals as they transition between job types, the direction of causality remains uncertain. Second, small sample sizes in some subgroups, particularly on-call and contract firm workers, limit statistical power and may mask important within-group variation.

A third limitation is the potential for omitted variable bias. Key psychological traits such as personality, risk tolerance, or preference for independence were not accounted for in my analysis but may strongly affect both employment choice and well-being. Finally, among other limitations, job satisfaction and happiness are measured using single-item indicators. While common in large surveys, such measures lack depth. Multi-dimensional scales could provide a more comprehensive picture of how workers experience their jobs and lives.

Directions for Future Research

Future studies should build on these findings by addressing limitations already discussed. First, longitudinal data is needed to better establish the causal direction between work

arrangements and subjective well-being. It remains unclear whether certain job structures increase happiness or whether individuals with higher baseline well-being and autonomy preferences are more likely to choose those roles.

Second, this study relies on assumptions about the defining features of each work type. For example, independent contracting is generally assumed to involve greater autonomy and flexibility, while on-call work is associated with instability and lack of control. However, these structural traits were not directly measured in the data. Future research should include variables that capture perceived job control, schedule predictability, autonomy, and access to benefits. Researchers can test whether the assumed characteristics of each employment type match up with how workers experience their jobs, and whether these features mediate the relationship between work arrangement and well-being. Similarly, it is worth looking at how workers evaluate their roles. Incorporating measures like learning opportunities, skill use, and chances for professional development could provide a clear picture of how job structure influences subjective experience. Lastly, lumping broad work categories together might hide key differences. For instance, independent contractors range from well-paid consultants to low-wage gig workers. Without more detailed occupational and industry data, we risk missing important inequalities within those groups.

8. Conclusion

This thesis investigates how different types of alternative work arrangements affect subjective well-being, specifically job satisfaction and happiness, compared to regular employment. In doing so, it responds to gaps in the literature by disaggregating alternative work into meaningful categories and examining their distinct effects. Much existing research focuses

narrowly on on-demand, low-wage gig work, overlooking the diversity of nonstandard employment. By broadening the analytical lens, this study offers a comprehensive understanding of how different work structures shape worker well-being.

The finding demonstrates that nonstandard work arrangements are not uniform in their effects on worker well-being. Independent contracting is associated with greater satisfaction and happiness, likely due to autonomy and greater alignment with personal values. In contrast, on-call work tends to have the opposite effect, with significantly lower job satisfaction and no corresponding benefits to happiness. Contract firm employment appears more neutral, yielding outcomes resembling those of regular employment.

Recognizing this heterogeneity is important. As the labor market continues to evolve, driven by technological growth, shifting worker preferences, and employer demand for flexibility, policy frameworks must also keep pace. Rather than treating all alternative work as either inherently risky or empowering, policy should focus on improving the actual conditions that define the work experience: autonomy, income stability, schedule predictability, and access to core protections.

Without this targeted approach, we risk designing policies that either miss their target or reinforce existing inequalities for those in non-standard roles. In economic terms, worker well-being is not just a social issue; it is foundational to productivity, human capital development, and the long-term strength of labor market institutions. As alternative work continues to grow, it is essential to build more inclusive communities and create an equitable economy.

Appendix

Appendix Table A1. Main Variable Definitions

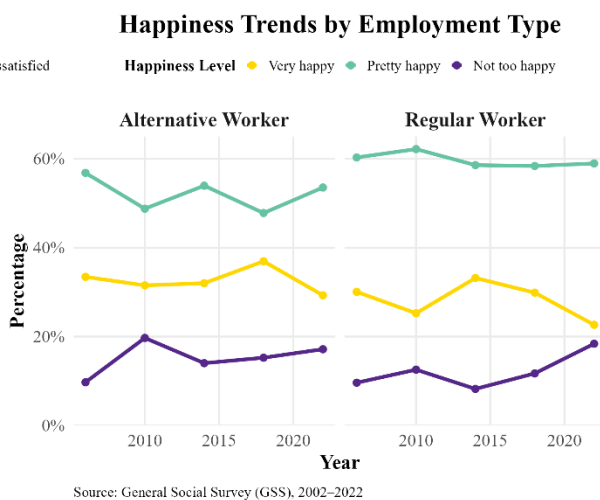
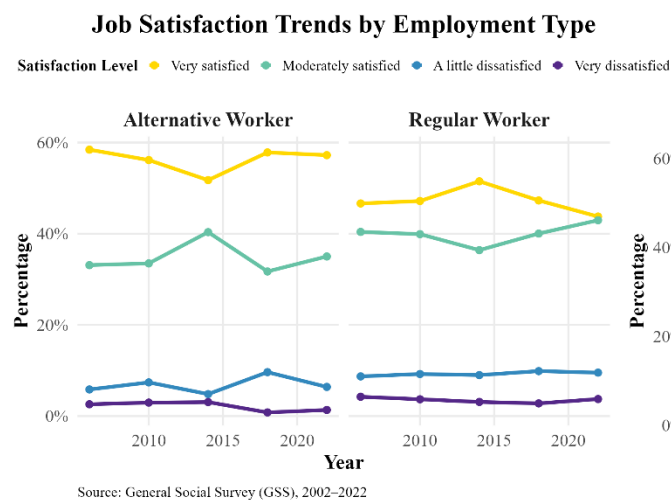
Variable	Definition	Response Options
happy	Overall happiness: “Taken all together, how would you say things are these days?”	Very Happy/ Pretty Happy/ Not too Happy
satjob	Job satisfaction: “On the whole, how satisfied are you with the work you do?”	Very satisfied / Moderately satisfied / A little dissatisfied / Very dissatisfied

Appendix Table A2. Control Variable Definitions

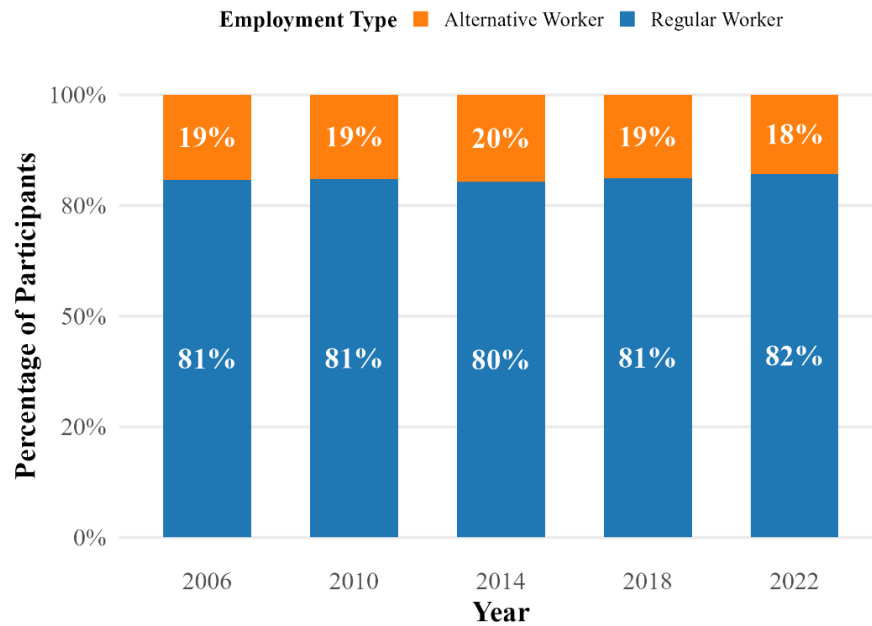
Variable	Definition
Age	Respondent’s Age
Sex	Respondent’s Sex
Race	Race of Respondent
Marital	Marital Status
Educ	Highest Year of School Completed
Region	Region of Interview
Wrktype	Work arrangement at main job Categories: Independent contractor/consultant/freelance worker, On-call, Temp agency, Contract work, Regular permanent employee <i>Note: The GSS does not provide explicit definitions for these categories. The following are based on definitions from the U.S. Bureau of Labor Statistics (BLS):</i> • Independent Contractor: Individuals who provide services to businesses or clients but are not classified as employees. They operate their own business (even if unincorporated) and may be identified as contractors, consultants, or freelance workers—regardless of whether they report as self-employed or wage and salary workers. • On-Call Worker: Workers who report to work only when needed.

	• Temporary Help Agency Worker: Individuals who are paid by a temporary help agency, whether or not their current job is considered temporary. • Contract Firm Worker: Workers employed by a contract company who are typically assigned to one client at a time and generally work on-site at the client's place of business.
indus10	Respondent's industry code (naics 2007)
occ10	Respondent's occupation code
mproff	A binary indicator denoting whether a respondent holds a managerial or professional occupation. <i>Classification:</i> Respondents are classified as managerial/professional if their occupation falls within the 2010 Census Occupation Codes 0010–3540. These job typically require a college education, involving high-level cognitive and technical skills, and often require formal credentials or licenses.
conrinc	Annual income in constant dollars

Appendix Figure A1. Trends in job satisfaction and happiness by employment type, 2006–2022.
Source: General Social Survey (GSS), 2002–2022.

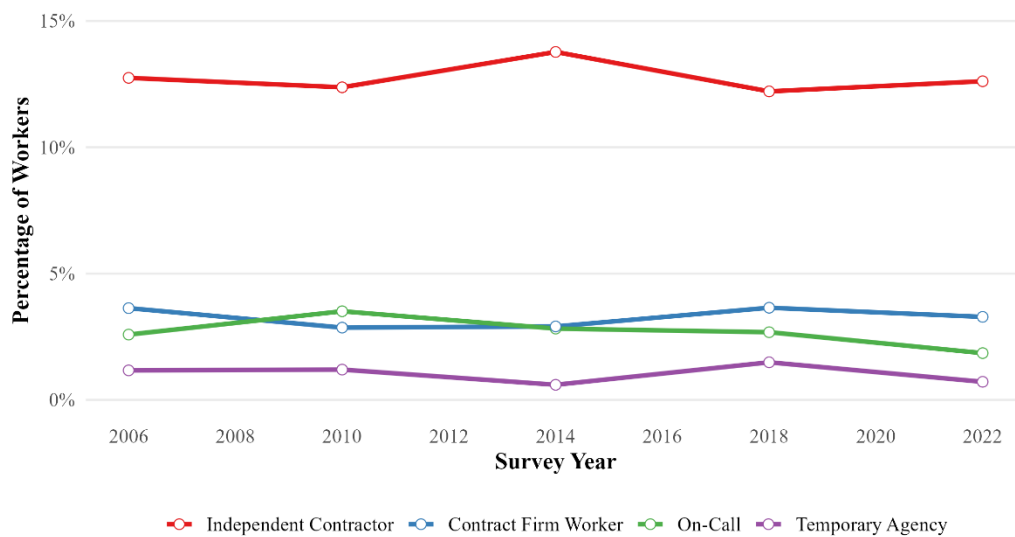


Appendix Figure A2: Share of regular and alternative workers in the U.S. labor force, 2006–2022.



Source: General Social Survey (GSS), 2002–2022

Appendix Figure A3. Share of U.S. labor force by detailed alternative work type, 2006–2022.



Source: General Social Survey (GSS), 2006–2022

Appendix Table A3: Summary Statistics by Worker Type

	Overall N = 6821	Regular Employee N = 5536	Independent Contractor N = 877	Contract Worker N = 228	On-Call Worker N = 180
General Happiness					
Not too happy	13%	12%	13%	20%***	16%
Pretty happy	58%	60%	51%***	52%**	58%
Very happy	29%	28%	35%***	28%	26%
Job Satisfaction					
Very dissatisfied	3.3%	3.6%	1.6%***	3.1%	3.3%
A little dissatisfied	8.8%	9.3%	5.1%***	7.5%	14%**
Moderately satisfied	39%	40%	29%***	46%	46%
Very satisfied	49%	47%	64%***	43%	37%**
Professional Occupation	40%	41%	37%***	29%***	26%***
Education (years)	14.22 (2.83)	14.31 (2.79)	14.18 (2.84)	13.38*** (3.11)	12.77*** (3.13)
Income (constant \$)	33,285 (34,813)	33,912 (33,276)	33,925 (43,657)	30,339*** (32,091)	14,588*** (30,447)
Age	43.06 (13.56)	42.30 (13.19)	47.87*** (14.17)	42.15 (13.16)	44.06* (16.90)
Gender %					
MALE	48%	46%	61%***	65%***	47%
FEMALE	52%	54%	39%***	35%***	53%
Weeks Worked Last Year	46.29 (11.83)	47.21 (10.88)	43.47*** (13.62)	44.42*** (12.55)	34.02*** (18.45)
Married	47%	47%	48%	43%	40%

Note: The sample consists primarily of regular employees (81.2%), followed by independent contractors (12.9%), contract firm workers (3.3%), and on-call workers (2.6%).

*** p<0.01, ** p<0.05, * p<0.1

Appendix Table A4: Industry Characteristics by Work Arrangement

Industry	Total Employed	Status on Employment			
		Regular Employee		Alternative Worker	
		% of total employed	Avg Sat Level	% of total employed	Avg Sat Level
Accommodation and Food Services	413	90.6	3.1	9.4	3.2
Administrative and Support and Waste Management and Remediation Services	241	67.6	3.2	32.4	3.3
Agriculture, Forestry, Fishing and Hunting	71	49.3	3.3	50.7	3.4
Arts, Entertainment, and Recreation	177	67.8	3.2	32.2	3.5
Construction	472	53.0	3.4	47.0	3.5
Educational Services	693	90.9	3.5	9.1	3.5
Finance and Insurance	346	86.1	3.3	13.9	3.5
Health Care and Social Assistance	1010	86.4	3.4	13.6	3.5
Information	112	89.3	3.2	10.7	3.3
Management of Companies and Enterprises	3	100.0	3.3	-	-
Manufacturing	660	89.5	3.2	10.5	3.4
Mining, Quarrying, and Oil and Gas Extraction	36	83.3	3.3	16.7	3.3
Other Services (except Public Administration)	389	62.2	3.4	37.8	3.5
Professional, Scientific, and Technical Services	436	74.5	3.3	25.5	3.5
Public Administration	462	93.9	3.4	6.1	3.5
Real Estate and Rental and Leasing	126	54.8	3.1	45.2	3.5
Retail Trade	692	88.7	3.1	11.3	3.3
Transportation and Warehousing	293	74.4	3.3	25.6	3.2
Utilities	63	87.3	3.5	12.7	3.6
Wholesale Trade	126	88.9	3.2	11.1	3.6

Note: Job satisfaction scale: 1= Very dissatisfied to 4= Very satisfied. Percentage represents proportion within each industry

Appendix Table A5: Fixed Effects Analysis

Region FE	General Happiness	Job Satisfaction
East South Atlantic	0.043 (0.035)	0.084** (0.043)
Middle Atlantic	0.028 (0.027)	0.001 (0.035)
Mountain	0.053* (0.030)	0.006 (0.039)
New England	-0.003 (0.038)	-0.021 (0.047)
Pacific	0.008 (0.026)	-0.001 (0.033)
South Atlantic	0.009 (0.024)	0.025 (0.030)
West North Central	-0.003 (0.033)	0.020 (0.041)
West South Central	0.062** (0.028)	0.122*** (0.035)

Year		
2010	-0.076*** (0.023)	-0.006 (0.030)
2014	0.035 (0.022)	0.032 (0.029)
2018	-0.006 (0.022)	0.006 (0.028)
2022	-0.147*** (0.021)	-0.047* (0.027)
Industry		
Admin & Waste Management	-0.044 (0.051)	0.083 (0.068)
Agriculture	0.110 (0.078)	0.092 (0.098)
Arts & Recreation	-0.030 (0.058)	0.081 (0.071)
Construction	0.015 (0.045)	0.222*** (0.053)
Education Services	-0.003 (0.041)	0.207*** (0.051)
Finance & Insurance	0.062 (0.046)	0.099* (0.057)
Health Care	0.003 (0.038)	0.202*** (0.047)
Information	0.027 (0.067)	0.019 (0.082)
Management	0.091 (0.234)	0.103 (0.313)
Manufacturing	-0.010 (0.040)	0.044 (0.051)
Mining	-0.119 (0.094)	0.010 (0.135)
Other Services (Except Public Administration)	0.045 (0.045)	0.184*** (0.056)
Professional Scientific and Tech Services	-0.025 (0.045)	0.046 (0.056)
Public Administration	0.004 (0.043)	0.173*** (0.055)
Real Estate	-0.067 (0.062)	-0.014 (0.076)
Retail Trade	-0.026 (0.039)	-0.015 (0.050)
Transportation	-0.022 (0.050)	0.047 (0.060)
Utilities	-0.070 (0.072)	0.306*** (0.084)
Wholesale Trade	0.045 (0.063)	0.009 (0.082)

Note: The reference group for categorical variables is set to the year 2006, the Accommodation industry, and the East North Central region. All reported coefficients reflect deviations from this baseline.

*** p<0.01, ** p<0.05, * p<0.1

Appendix Table A6: Robustness Check – Clustering Standard Errors by Industry

	General Happiness (1)	Job Satisfaction (2)
Employment Relationships		
Independent contractor/ consultant/ freelance worker	0.072*** (0.019)	0.217*** (0.022)
On-call worker	-0.010 (0.043)	-0.124*** (0.043)
Work for contractors who provides services	-0.041 (0.025)	-0.016 (0.037)
Covariates		
Managerial/Professional	0.0352 (0.025)	0.095** (0.032)
Education (years)	0.007* (0.004)	-0.001 (0.004)
Income (thousands)	0.0012*** (0.000)	0.001*** (0.000)
Age	-0.002** (0.001)	0.006*** (0.001)
Married	0.303*** (0.015)	0.095*** (0.011)
Female	0.039** (0.017)	0.014 (0.016)
Black	0.001 (0.017)	-0.054** (0.025)
Other Race	-0.000 (0.025)	0.034 (0.028)
Constant	1.941*** (0.065)	2.794*** (0.065)
Fixed Effects		
Survey FE	Yes	Yes
Region FE	Yes	Yes
Industry FE	Yes	Yes
Year FE	Yes	Yes
Observations	6,821	6,821

Note: Standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Appendix Table A7: Regression Coefficients by Industry and Employment Type

Industry	N	Happiness Coefficients			Job Satisfaction Coefficient		
		Indep. Contractor	On-call	Contract Firm	Indep Contractor	On-call Worker	Contract Firm
Accommodation and Food	413	0.202 (0.126)	0.210 (0.195)	-0.056 (0.378)	0.369*** (0.133)	-0.426*** (0.108)	-0.677 (0.576)
Admin & Support	241	-0.007 (0.106)	0.136 (0.254)	-0.016 (0.137)	0.010 (0.143)	-0.177 (0.253)	0.149 (0.164)
Arts & Entertainment	177	0.077 (0.124)	0.383* (0.168)	0.073 (0.499)	0.274** (0.140)	-0.056 (0.287)	-0.298 (1.076)
Construction	472	0.035 (0.075)	-0.009 (0.156)	-0.075 (0.088)	0.292*** (0.069)	-0.328* (0.187)	0.002 (0.096)
Educational Services	693	0.039 (0.110)	0.048 (0.108)	-0.186 (0.189)	0.337*** (0.100)	-0.133 (0.194)	-0.199* (0.132)
Finance & Insurance	346	0.003 (0.085)	-0.081 (0.118)	-0.156 (0.420)	0.194 (0.120)	-0.286* (0.168)	0.084 (0.236)
Health Care	1010	0.039 (0.075)	0.058 (0.109)	-0.007 (0.131)	0.216*** (0.077)	-0.185 (0.175)	0.094 (0.117)
Information	112	0.313 (0.283)	0.776 (0.397)	-0.092 (0.514)	0.408 (0.287)	0.488 (0.391)	-0.279 (0.547)
Manufacturing	600	0.219* (0.120)	0.355 (0.205)	0.040 (0.124)	0.267*** (0.102)	0.011 (0.159)	0.024 (0.135)
Other Services	389	0.006 (0.071)	-0.121 (0.156)	-0.212 (0.267)	0.197** (0.091)	-0.112 (0.201)	0.289* (0.173)
Professional Services	436	-0.015 (0.077)	-0.620*** (0.162)	0.147 (0.132)	0.250*** (0.091)	0.001 (0.215)	-0.199 (0.230)
Public Admin	462	0.084 (0.173)	-0.382** (0.183)	0.020 (0.136)	-0.118 (0.252)	0.631*** (0.099)	0.016 (0.178)
Real Estate	126	0.166 (0.126)	0.422 (0.414)	0.112 (0.295)	0.259 (0.161)	0.795*** (0.276)	1.404*** (0.427)
Retail	692	-0.004 (0.086)	0.028 (0.141)	-.127 (0.215)	0.323*** (0.104)	-0.062 (0.235)	-0.353 (0.309)
Transportation	293	-0.122 (0.103)	-0.062 (0.194)	-.017 (0.172)	-0.000 (0.135)	-0.123 (0.174)	-0.051 (0.210)
Wholesale	126	.035 (0.180)	-0.254 (0.337)	n/a	0.340 (0.235)	0.262 (0.600)	n/a

Note: Industries with insufficient representation—agriculture (n = 71), management (n = 3), mining (n = 36), and utilities (n = 63) are excluded from the analysis.

*** p<0.01, ** p<0.05, * p<0.1

Appendix Table A8: Ordered Logit Models: Happiness and Job Satisfaction and Quadratic Controls

	Logit: Happiness	Logit: Job Satisfaction	OLS: Happiness (Quad)	OLS: Job Sat. (Quad)
Employment Relationships				
Independent contractors	0.258*** (0.069)	0.634*** (0.073)	0.518** (0.218)	0.742** (0.274)
On-call worker	-0.027 (0.144)	-0.343*** (0.097)	-0.203 (0.464)	0.287 (0.408)
Work for contractors who provides services	-0.129 (0.090)	-0.090 (0.085)	0.292 (0.373)	-0.020 (0.412)
Covariates				
Income (thousands)	0.004*** (0.001)	0.004*** (0.001)	0.002** (0.001)	0.001 (0.001)
Income ²			0.000 (0.000)	-0.000 (0.000)
Age	-0.006** (0.003)	0.016*** (0.002)	-0.011** (0.004)	-0.005 (0.005)
Age ²			0.0001 (0.000)	0.000 (0.000)
Interaction: EmpType x Age				
Independent Contractor × Age			-0.015 (.009)	-0.020* (0.011)
On-call x Age			0.010 (0.021)	-0.032 (0.019)
Contractor Worker x Age			-0.016 (0.020)	0.000 (0.020)
Interaction: EmpTye x Age²				
Independent Contractor × Age ²			-0.015 (0.009)	0.000 (0.000)
On-call x Age ²			0.010 (0.021)	0.0004* (0.000)
Contractor Worker x Age ²			-0.016 (0.020)	0.000 (0.000)
Interaction: EmpType x Income				
Independent Contractor × Income			-0.002 (0.002)	-0.001 (0.001)
On-call x Income			-0.002 (0.004)	0.016*** (0.004)
Contractor Worker x Income			0.004 (0.004)	-0.005 (0.004)
Interaction: EmpTye x Income²				
Independent Contractor × Income ²			0.000 (0.000)	0.000 (0.000)
On-call x Income ²			0.000 (0.000)	-0.0001*** (0.000)
Contractor Worker x Income ²			0.000 (0.000)	0.000 (0.000)
Cut 1				
	-1.323*** (0.204)	-2.144*** (0.190)		
Cut 2				
	1.708*** (0.240)	-0.721*** (0.174)		
Cut 3				
		1.426*** (0.164)		
Constant			2.110*** (0.111)	3.002*** (0.112)
Observations	6,821	6,821	6,821	6,821

Note: Standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Appendix Table A9: Interaction Effects on Happiness and Job Satisfaction

	EmpType x Sex		EmpType x Married		EmpType x MPROFF	
	Happiness	Job Satisfaction	Happiness	Job Satisfaction	Happiness	Job Satisfaction
Employment Relationship						
Independent contractor	0.058** (0.021)	0.239*** (0.024)	0.102*** (0.027)	0.226*** (0.029)	0.088** (0.037)	0.213*** (0.036)
On-call worker	-0.025 (0.055)	-0.061 (0.073)	-0.059 (0.055)	-0.149** (0.064)	-0.020 (0.046)	-0.147** (0.052)
Contract worker	-0.046 (0.037)	0.028 (0.066)	-0.099** (0.038)	-0.026 (0.066)	-0.036 (0.037)	0.024 (0.051)
Sex						
Female	0.034** (0.014)	0.026* (0.015)	0.039** (0.016)	0.014 (0.016)	0.024 (0.019)	0.010 (0.016)
Independent contractor × Female	0.031 (0.039)	-0.049 (0.050)	—	—	—	—
On-call × Female	0.029 (0.074)	-0.117 (0.074)	—	—	—	—
Contractor × Female	0.009 (0.065)	-0.116 (0.106)	—	—	—	—
Marriage						
Married	0.303*** (0.015)	0.095*** (0.011)	—	—	—	—
Independent contractor × Married=1	—	—	-0.062 (0.038)	-0.020 (0.034)	—	—
On-call × Married=1	—	—	0.124 (0.088)	0.063 (0.077)	—	—
Contractor × Married=1	—	—	0.136 (0.082)	0.024 (0.106)	—	—
Professional Status						
MPROFF = 1	—	—	—	—	0.064** (0.025)	0.102** (0.034)
Independent contractor × MPROFF=1	—	—	—	—	-0.065 (0.054)	-0.005 (0.053)
On-call × MPROFF=1	—	—	—	—	0.010 (0.079)	0.083 (0.063)
Contractor × MPROFF=1	—	—	—	—	-0.050 (0.103)	-0.146 (0.095)
Constant	1.943*** (0.064)	2.787*** (0.064)	1.942*** (0.065)	2.795*** (0.065)	1.993*** (0.063)	2.810*** (0.064)
Observations	6,821	6,821	6,821	6,821	6,821	6,821

Note: Standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

Appendix Table A10: Fully Adjusted Regression Models with Year Interactions: Happiness and Job Satisfaction by Employment Type (2006–2022)

	Happiness	Job Satisfaction
Employment Relationship		
Independent contractor	0.095* (0.049)	0.232*** (0.056)
On-call worker	-0.159* (0.087)	-0.147 (0.107)
Work for contractor	0.001 (0.053)	0.057 (0.083)
Year		
2010	-0.071*** (0.016)	0.002 (0.034)
2014	0.055*** (0.018)	0.061** (0.027)
2018	-0.010 (0.018)	0.007 (0.034)
2022	-0.159*** (0.020)	-0.058 (0.038)
Emp Type x Year Interaction Terms		
IC × 2010	-0.062 (0.057)	-0.023 (0.096)
IC × 2014	-0.099 (0.059)	-0.145 (0.095)
IC × 2018	-0.020 (0.093)	-0.030 (0.081)
IC × 2022	0.029 (0.075)	0.083 (0.078)
On-call × 2010	0.232 (0.210)	0.005 (0.191)
On-call × 2014	0.070 (0.116)	-0.171 (0.258)
On-call × 2018	0.254* (0.132)	0.202 (0.135)
On-call × 2022	0.205 (0.119)	0.064 (0.157)
Contractor × 2010	-0.116 (0.122)	-0.157 (0.137)
Contractor × 2014	-0.302* (0.158)	-0.134 (0.151)
Contractor × 2018	-0.010 (0.074)	-0.084 (0.102)
Contractor × 2022	0.090 (0.078)	-0.052 (0.147)
Covariates		
MPROFF	0.036 (0.025)	0.096*** (0.032)
Education (Years)	0.007* (0.004)	0.002 (0.004)

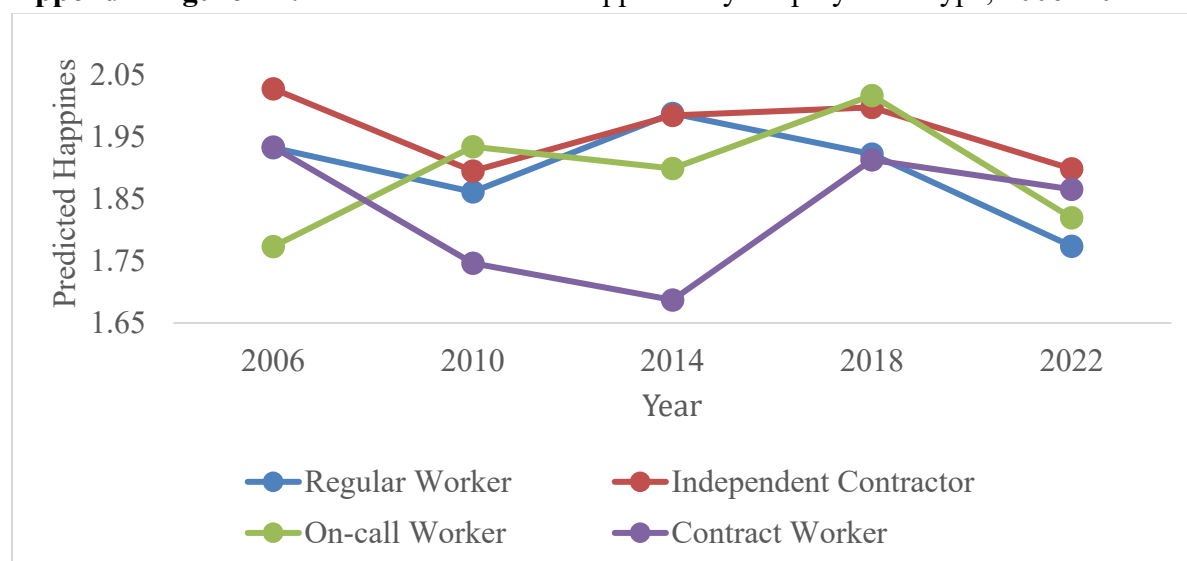
Income (k)	0.001*** (0.000)	0.001*** (0.000)
Age	-0.002* (0.001)	0.006*** (0.001)
Married	0.303*** (0.015)	0.095*** (0.011)
Female	0.040** (0.017)	0.015 (0.016)
Black	0.000 (0.016)	-0.055** (0.025)
Other race	-0.002 (0.025)	0.034 (0.028)
Constant	1.933*** (0.066)	2.785*** (0.066)

Note: Standard errors in parentheses *** p<0.01, ** p<0.05, * p<0.1

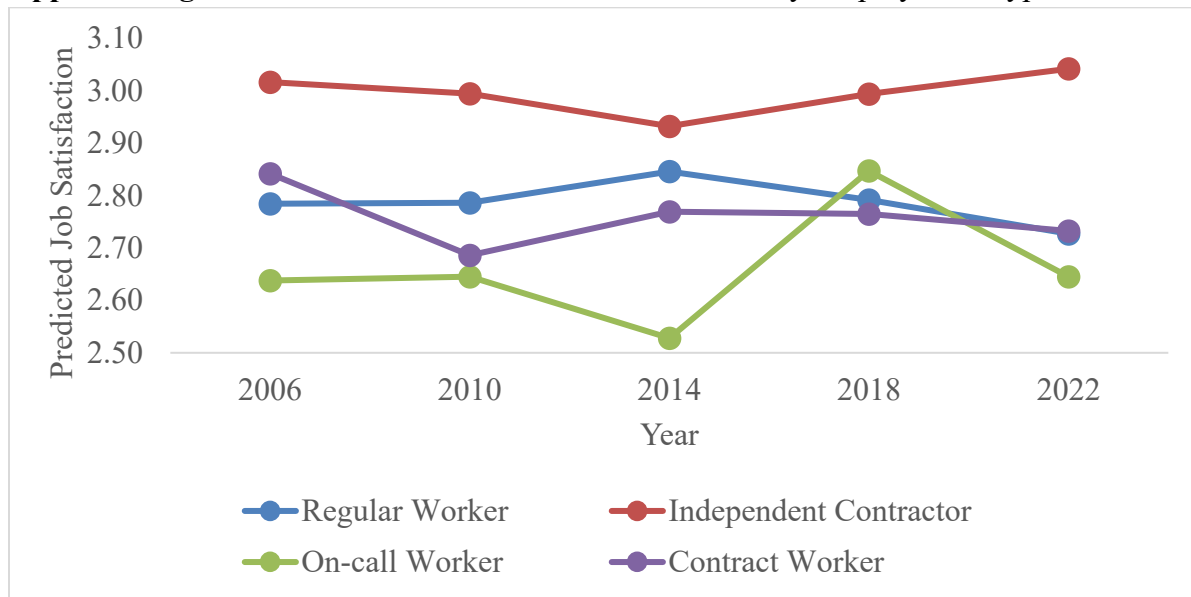
Appendix Table A12: Predicted Happiness and Job Satisfaction by Employment Type and Year (2006–2022)

Year	Predicted Happiness				Predicted Job Satisfaction			
	Regular	Indep.	On-call	Contract	Regular	Indep.	On-call	Contract
2006	1.933	2.027	1.773	1.933	2.785	3.016	2.638	2.842
2010	1.861	1.895	1.934	1.747	2.786	2.995	2.645	2.686
2014	1.988	1.985	1.899	1.687	2.846	2.932	2.528	2.769
2018	1.923	1.998	2.017	1.913	2.792	2.994	2.847	2.765
2022	1.774	1.898	1.82	1.866	2.727	3.042	2.645	2.733

Appendix Figure A4. Trends in Predicted Happiness by Employment Type, 2006–2022.



Appendix Figure A5. Trends in Predicted Job Satisfaction by Employment Type, 2006–2022.



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